Manan Agrawal

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SUMMARY

AI Engineer with 3+ years of experience building production AI systems, Machine Learning solutions, and Generative AI applications. Delivered enterprise LLM, Agentic AI, and Retrieval Augmented Generation solutions processing 500K+ records daily while reducing manual effort by up to 60%. Passionate about building reliable AI software that transforms business problems into measurable outcomes using modern AI and cloud technologies.

WORK EXPERIENCE

Martinrea International US Inc. | ***Research and Development AI Associate*** **Aug 2025 - May 2026**

- Architected production machine learning pipelines using Python, Azure SQL, and enterprise data engineering workflows to process over 500K innovation records daily, enabling intelligent idea prioritization and significantly improving business decision making across multiple departments.
- Developed predictive AI models for ROI estimation, implementation success, anomaly detection, and demand forecasting that improved planning accuracy by 35%, accelerated investment decisions, and reduced manual evaluation effort for cross functional business teams.
- Engineered scalable ETL pipelines, automated feature engineering workflows, and real time analytics dashboards that reduced reporting latency by 25%, increased data reliability, and provided executives with actionable AI driven operational insights.
- Collaborated with product managers, engineering teams, and business stakeholders to deploy production ready AI solutions from experimentation through enterprise adoption, translating machine learning outputs into measurable commercial value and organization wide process improvements.

Honeywell | ***Generative Ai Associate*** **Jan 2025 - May 2025**

- Designed and deployed a production Retrieval Augmented Generation platform leveraging OpenAI models, semantic embeddings, vector search, and FastAPI services to automate enterprise service ticket classification with over 95% prediction accuracy across customer support operations.
- Built scalable LLM powered microservices capable of processing more than 1,000 production requests daily while implementing retrieval pipelines, prompt optimization strategies, and human in the loop validation to continuously improve model quality and reliability.
- Established comprehensive evaluation frameworks, automated monitoring pipelines, and AI performance metrics that reduced manual ticket processing effort by 60%, improved response consistency, and accelerated issue resolution across enterprise workflows.
- Partnered closely with software engineers, product owners, and business stakeholders to productionize GenAI applications, ensuring secure deployment, governance compliance, seamless API integration, and successful enterprise adoption of AI powered automation solutions.

SS Infotech | ***Machine Learning Engineer*** **Jul 2022 - Jun 2024**

- Designed and deployed end to end machine learning solutions for customer segmentation, predictive analytics, forecasting, and intelligent business automation using Python, SQL, and Scikit learn, enabling data driven decision making across multiple client engagements.
- Built scalable ETL pipelines, automated model training workflows, and feature engineering frameworks that processed high volume enterprise datasets, improved model reproducibility, and accelerated deployment of production ready machine learning solutions.
- Optimized supervised learning models through advanced feature selection, hyperparameter tuning, cross validation, and model evaluation techniques, improving prediction accuracy while reducing operational overhead for business critical analytics workflows.
- Collaborated directly with cross functional stakeholders to understand business challenges, translate requirements into scalable AI solutions, and deliver analytics dashboards, predictive models, and automation systems that generated measurable productivity improvements.

TECHNICAL SKILLS

- **Languages:** Python, Java, C++, SQL, JavaScript, TypeScript
- **AI & ML:** Machine Learning, Deep Learning, Generative AI, LLMs, Agentic AI, Multi Agent Systems, Retrieval Augmented Generation, Prompt Engineering, NLP, Computer Vision
- **Frameworks:** PyTorch, TensorFlow, Scikit learn, LangChain, LangGraph, LlamaIndex, Hugging Face, FastAPI, MLflow
- **Data & Backend:** PostgreSQL, Azure SQL, MySQL, FAISS, Pinecone, Weaviate, REST APIs, Async Programming, Feature Engineering, ETL Pipelines
- **Cloud & Devops:** AWS, Azure, Docker, Kubernetes, Git, CI/CD, Model Evaluation, Model Monitoring

PROJECTS

AlphaForage | AI Powered Multi Agent Financial Research Platform

- Architected a production grade multi agent financial intelligence platform using LangGraph, FastAPI, PostgreSQL, React, and OpenAI APIs, enabling autonomous reasoning across valuation, technical analysis, macroeconomics, sentiment analysis, and portfolio risk assessment.
- Developed advanced Retrieval Augmented Generation pipelines with hybrid semantic retrieval, vector databases, confidence calibration, citation grounded responses, and automated evaluation workflows, significantly improving factual accuracy and explainability of AI generated research.
- Engineered scalable backend infrastructure featuring streaming APIs, authentication, asynchronous processing, workflow orchestration, and intelligent agent collaboration, delivering enterprise ready AI capabilities for real time financial research and investment analysis.

Enterprise AI Document Intelligence Platform

- Built an enterprise AI document intelligence platform using LlamaIndex, FastAPI, OCR, embeddings, and Retrieval Augmented Generation to automate large scale document understanding, semantic search, and intelligent knowledge extraction across unstructured business data.
- Designed evaluation pipelines, prompt engineering strategies, retrieval optimization techniques, and vector search workflows that significantly improved response accuracy while reducing manual document review effort across enterprise knowledge management processes.
- Implemented scalable backend APIs, asynchronous processing pipelines, and production ready AI architecture supporting secure document ingestion, intelligent information retrieval, and conversational enterprise assistants deployed on modern cloud infrastructure.

HONORS AND PUBLICATIONS

- IEEE Peer Reviewer for machine learning and AI conferences and journals
- Published 15+ research papers in ML, AI, and applied statistics
- Govt. of India Design Patent No 386153 001
- Winner, Most Complete Solution, LPL Financial Hackathon (Alpha: AI platform)
- Invited Guest Speaker, Johnson C. Smith University, Charlotte

EDUCATION

UNC Charlotte | ***MS, Computer Engineering*** **Aug 2024 - May 2026**

RCOEM, Nagpur University | ***BTech, Electronics Engineering*** **Jul 2020 - Jun 2024**